

Comparative Evaluation of Classical Geometric Methods for Steppable-Surface Segmentation in Simulated Staircase Point Clouds

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Abstract—Reliable identification of steppable surfaces is a prerequisite for stair-climbing and stair-following robots. This paper presents a controlled, fully labeled benchmark for steppable-surface segmentation built from a parametric mathematical model of a four-step staircase point cloud with LiDAR-style noise and outliers, and a systematic comparison of four classical geometric segmentation methods: orientation-gated RANSAC plane fitting, PCA-based normal analysis, orientation-aware DBSCAN clustering with slope analysis, and height-histogram analysis. On a cloud of 18,333 points, the DBSCAN-with-slope method attained the highest tread-class scores ($F1 = 0.943$, $IoU = 0.892$), followed closely by PCA normal analysis ($F1 = 0.927$). Plain RANSAC was the least precise because horizontal plane hypotheses absorb riser points at tread heights. A noise-sensitivity study shows the tread-class $F1$ degrading from 0.956 to 0.543 as the height noise σ_z grows from 0.01 m to 0.07 m. We discuss the safety implications of these error modes for robotic stair traversal and argue for ensemble fusion with safety margins. The benchmark, code, and equations are released to support reproducible comparison.

Index Terms—point cloud, segmentation, RANSAC, PCA, DBSCAN, staircase, mobile robot, traversability, LiDAR

I. INTRODUCTION

Robots that climb or follow staircases and assistive devices for the visually impaired must reason about three-dimensional structure to act safely. Legged and wheeled platforms increasingly rely on egocentric depth sensing to negotiate stairs and other discontinuous terrain [1]–[3], while shared-autonomy systems demand reliable geometric cues before a foot or wheel is committed to a surface [4]. In all of these settings the central perceptual question is the same: which parts of an observed point cloud constitute a *tread*, the steppable horizontal surface on which weight can be safely placed, as opposed to a near-vertical *riser* or spurious edge and outlier returns.

This problem is safety-critical. A tread mislabeled as non-steppable merely sacrifices an option, but a riser or edge mislabeled as steppable invites a foothold onto a vertical face, producing slips, stumbles, or falls. The same hazard motivates stair perception for the visually impaired, where undetected steps are a recognised fall hazard that motivates dedicated stair-detection systems [5]. Reliable tread detection therefore

requires not only high recall but also high precision on the steppable class.

Recent work approaches stair and terrain perception with deep neural networks that learn point-wise semantics directly from raw clouds [5], [6]. Such methods are powerful but data hungry: they require large volumes of densely labeled three-dimensional scans, which are costly to acquire for the specific geometry of stairs, and they offer limited interpretability. Equally problematic, the stair tread sub-problem lacks controlled, reproducible benchmarks with exact ground-truth labels, so reported results are difficult to compare across studies. Classical geometric methods, by contrast, are fast, transparent, and label-free, yet despite their maturity in plane segmentation and traversability estimation [7], [8] they remain under-compared on the staircase setting under matched conditions.

We address this gap with a controlled, fully reproducible study of classical geometric segmentation for steppable-surface detection on staircases. Our contributions are as follows:

- A parametric, fully labeled staircase point-cloud generator that emits exact tread, riser, and other labels while modeling LiDAR range jitter and outliers, enabling reproducible evaluation.
- A unified three-class problem formulation together with the implementation of four classical methods under a common orientation-aware pipeline: RANSAC plane fitting, PCA normal analysis, DBSCAN with slope confirmation, and a height-histogram level detector.
- A quantitative comparison of all four methods on the steppable (tread) class, accompanied by a controlled noise-sensitivity study that characterizes degradation as height noise grows.
- A safety-oriented discussion of the results for robotic stair traversal, relating precision and intersection-over-union to the risk of unsafe footholds.

The remainder of this paper is organized as follows: Section II reviews related work, Section III presents the proposed data generator and the four segmentation methods, Section IV

reports the comparative evaluation and noise study, and Section V concludes.

II. RELATED WORK

Detecting steppable surfaces in staircase point clouds draws on several threads of point-cloud processing. We organise the literature thematically and contrast each line of work with the requirements of our problem: fast, interpretable, and label-free segmentation of treads and risers.

A. Geometric plane segmentation

The dominant paradigm for extracting planar structure from unorganised clouds is Random Sample Consensus (RANSAC). Li *et al.* [9] improve standard RANSAC for indoor and urban plane segmentation by guiding sampling with Normal Distribution Transformation cells, mitigating the spurious planes that plain RANSAC fits across disjoint surfaces. RANSAC plane fitting has likewise been applied to RGB-D object recognition [10] and embedded in real-time planar LiDAR SLAM [11], where planes serve as map primitives. These methods are efficient and interpretable, but they treat planes as generic primitives without an orientation prior; on a staircase the globally dominant plane is the oblique ramp spanning all steps, so unconstrained fitting absorbs riser points into tread planes. This observation motivates the orientation gate in our pipeline.

B. Normal estimation and surface analysis

A complementary route reasons about local surface geometry rather than global primitives. Huang *et al.* [12] consolidate unorganised clouds and estimate reliable normals for surface reconstruction, while Castillo *et al.* [13] recover normals through constrained nonlinear least squares for joint segmentation and denoising. Per-point normals are attractive for stairs because tread and riser differ sharply in orientation, yet normal estimation alone yields no grouping into coherent surfaces and is sensitive to neighbourhood scale under noise. Our PCA-normal baseline builds directly on this principle, classifying points by the tilt of the smallest-eigenvector normal relative to the vertical axis.

C. Clustering and region growing

To recover surfaces rather than isolated points, clustering and region-growing methods aggregate neighbours by geometric similarity. Density-based clustering (DBSCAN) has been used to identify and localise 3D weld seams robustly in the presence of outliers [14]; Euclidean clustering supports object detection from fused sparse clouds and imagery [15]; and supervoxel-based region growing produces compact, boundary-respecting segments [16]. These approaches naturally reject outliers as noise and form one cluster per physical surface, which suits the discrete tread/riser structure of stairs. They do not, however, encode the horizontal-versus-vertical distinction we require, so we precede spatial clustering with an orientation split and confirm each cluster by its slope.

D. Ground and LiDAR segmentation

Separating traversable ground from obstacles is a long-studied special case of surface segmentation. The survey by Gomes *et al.* [8] catalogues ground-segmentation methods for automotive LiDAR and highlights the recurring trade-off between geometric methods that are fast and transparent and learned methods that are accurate but opaque. Stairs are a structured instance of this ground-versus-non-ground problem, where each tread is a small, elevated ground patch rather than a single dominant plane.

E. Learning-based segmentation

Deep networks have reshaped point-cloud semantic segmentation since PointNet [6] demonstrated direct learning on raw point sets, and recent surveys document the breadth of this progress [17]. Because dense labels are costly, weakly supervised [18], few-shot [19], and prototype-based few/zero-shot [20] formulations aim to reduce annotation demand. Despite strong accuracy, these models remain data-hungry, depend on representative training distributions, and offer limited interpretability and reproducibility for a safety-critical subproblem such as foothold detection. This data and transparency cost is precisely what motivates revisiting classical geometry for stairs.

F. Robotic stair and terrain perception

In robotics, stair and terrain perception is tightly coupled to control. Matsumura *et al.* [5] detect stairs from 3D point clouds with deep learning to prevent falls, while operator-assisted stair traversal has been demonstrated in field systems [4]. Legged platforms rely on elevation maps and adaptive planning [2], egocentric vision [1], and stereo-based navigation over rough terrain [3], and traversability itself can be learned from simulation [7]. These systems show that reliable steppable-surface detection is a prerequisite for safe motion, yet they are typically evaluated end-to-end on bespoke platforms, leaving the underlying segmentation step without an isolated, comparable benchmark.

G. Gap addressed by this work

Across these threads, geometric methods are fast and interpretable but rarely compared head-to-head on stairs, while learned methods require labelled data unavailable for this specific task. No prior study offers a controlled, fully labelled staircase benchmark on which classical segmenters can be measured against a common ground truth. We fill this gap with a parametric, fully annotated point-cloud generator and an apples-to-apples comparison of four classical methods for steppable-surface detection.

III. PROPOSED METHOD

This section formalises the steppable-surface segmentation problem, describes a parametric generator that produces a fully labelled staircase point cloud with realistic sensor artefacts, and details four classical geometric methods that we implement and compare under an identical 3-class formulation.

Algorithm 1 Orientation-aware segmentation (generic)

- 1: estimate per-point normals $\hat{\mathbf{n}}_p$ via PCA over k neighbours (4)
 - 2: split points into horizontal / vertical orientation groups
 - 3: cluster or fit planes within each group
 - 4: confirm each region by its plane tilt $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}|$
 - 5: label tread / riser / other
-

A. Problem Formulation and Class Scheme

We treat staircase perception as a point-wise classification problem. Given an unorganised cloud $\mathcal{P} = \{\mathbf{x}_i\}_{i=1}^M$ with $\mathbf{x}_i \in \mathbb{R}^3$, the goal is to assign every point a label $\ell_i \in \{0, 1, 2\}$, where label 0 denotes a *tread* (a steppable, nominally horizontal surface), label 1 denotes a *riser* (the vertical face connecting consecutive treads), and label 2 denotes *other* (structural edges, sensor noise, and outliers). The tread class is the primary class of interest, because a robot or assistive device plans footholds on tread surfaces; consequently we evaluate it in a one-vs-rest setting while still reporting overall 3-class accuracy.

The decisive geometric cue separating the first two classes is surface orientation relative to the vertical axis. Let $\hat{\mathbf{n}}$ be the unit surface normal at a point and $\hat{\mathbf{z}}$ the world vertical. A tread satisfies $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| \rightarrow 1$ (its normal is nearly parallel to $\hat{\mathbf{z}}$), whereas a riser satisfies $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| \rightarrow 0$ (its normal is nearly orthogonal to $\hat{\mathbf{z}}$). Every method below exploits this orientation contrast, either explicitly through a normal-based gate or implicitly through the height structure of the cloud. Treating orientation as the organising principle yields a unified pipeline, summarised in Algorithm 1, into which each of the four methods plugs at the clustering or plane-fitting stage.

B. Parametric Data Generation

To obtain a controlled and fully labelled benchmark, we synthesise the staircase analytically from physical parameters: width $w = 2.5$ m, tread depth $d = 0.30$ m, step height $h = 0.18$ m, and $N = 4$ steps. For step index $i \in \{0, \dots, N-1\}$, tread points are sampled uniformly over the horizontal extent of step i and placed at height ih with additive Gaussian height noise, as in (1):

$$\begin{cases} x \sim \mathcal{U}(0, w) \\ y \sim \mathcal{U}(id, (i+1)d) \\ z = ih + \epsilon_z, \quad \epsilon_z \sim \mathcal{N}(0, \sigma_z^2) \end{cases} \quad (1)$$

$$\begin{cases} x \sim \mathcal{U}(0, w) \\ y = id + \epsilon_y, \quad \epsilon_y \sim \mathcal{N}(0, \sigma_y^2) \\ z \sim \mathcal{U}((i-1)h, ih) \end{cases} \quad (2)$$

Riser points, generated by (2) for $i \in \{1, \dots, N-1\}$ (the vertical face rising to tread i), share the same lateral spread but lie on a vertical plane at depth id with small along-depth jitter $\sigma_y = 0.02$ m, their height sweeping uniformly across one step rise. The nominal height noise is $\sigma_z = 0.02$ m, a value we later vary to probe sensitivity. On top of this structured

cloud we inject realism in two ways. First, approximately 4% of the points are replaced by uniformly distributed outliers spanning the staircase bounding volume, mimicking spurious returns and edge artefacts; these are labelled as *other*. Second, every retained point receives an isotropic LiDAR-like jitter drawn from $\mathcal{N}(0, 0.005^2)$ on each coordinate, approximating the few-millimetre range noise typical of indoor depth sensors. We treat this value as a nominal modelling choice rather than a calibrated sensor model, and probe its principal axis (σ_z) directly in Section IV. The resulting cloud contains 18,333 points, partitioned as 12,800 tread, 4,800 riser, and 733 other. Because the generator emits per-point labels by construction, it provides exact ground truth for both training-free evaluation and the noise sweep, removing the manual-annotation bottleneck that hampers learned segmenters.

C. RANSAC Plane Fitting

The first method fits planes by Random Sample Consensus. A candidate plane with unit normal $\hat{\mathbf{n}}$ and offset b scores points by their orthogonal distance, and the inlier set $\mathcal{I}$ collects points closer than a tolerance τ :

$$d(\mathbf{x}) = |\hat{\mathbf{n}}^\top \mathbf{x} + b|, \quad \mathcal{I} = \{\mathbf{x} : d(\mathbf{x}) < \tau\} \quad (3)$$

A naive single fit fails on stairs: the dominant plane that maximises inliers is the oblique surface spanning the entire staircase, whose normal has $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| = d/\sqrt{d^2 + h^2} \approx 0.86$. RANSAC therefore returns a spurious “ramp” that absorbs both treads and risers into one steppable region. To prevent this, we apply a two-phase, orientation-gated scheme: we first extract horizontal planes (treads) by accepting only fits whose normal satisfies $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| \geq 0.93$, then re-run RANSAC on the residual points to extract vertical planes (risers) with $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| \leq 0.25$. We set $\tau = 0.035$ m and impose minimum inlier counts of 2,000 points for tread planes and 300 for riser planes, which suppresses thin riser slices from contaminating tread fits at the cost of occasionally rejecting sparse vertical faces. This orientation-aware variant follows the spirit of structure-guided RANSAC refinements [9], [10].

D. PCA Normal Analysis

The second method classifies each point purely from its local surface orientation. For point p we gather its k nearest neighbours $\mathcal{N}_k(p)$, form the local covariance matrix $\mathbf{C}_p$, and take the eigenvector associated with its smallest eigenvalue as the surface normal:

$$\mathbf{C}_p = \frac{1}{k} \sum_{j \in \mathcal{N}_k(p)} (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top, \quad \hat{\mathbf{n}}_p = \mathbf{v}_{\min}(\mathbf{C}_p) \quad (4)$$

The smallest-eigenvalue direction is the axis of least point spread, which for a locally planar patch coincides with its normal [12], [13]. With $k = 50$ the estimate is stable against the injected jitter while remaining local enough to respect tread–riser boundaries. We classify by the verticality of the estimated normal: a point is labelled tread when $|\hat{\mathbf{n}}_p^\top \hat{\mathbf{z}}| \geq 0.80$

and riser when $|\hat{\mathbf{n}}_p^\top \hat{\mathbf{z}}| \leq 0.50$; intermediate values, which arise mostly at edges and from outliers, fall to the *other* class. This decision is per-point and parameter-light, requiring no plane-fitting iterations. These PCA thresholds (0.80/0.50) are deliberately looser than the RANSAC orientation gate (0.93/0.25): RANSAC tests a global plane whose aggregate normal is precise, whereas per-point PCA normals are noisier and require a wider acceptance band.

E. DBSCAN with Slope Confirmation

The third method couples orientation reasoning with spatial clustering. We first pre-split the cloud into a horizontal group and a vertical group using the same normal-verticality criterion as above. Within each group we then run density-based spatial clustering (DBSCAN) with neighbourhood radius $\varepsilon = 0.08$ m, so that each physically separate surface forms its own connected cluster while sparse outliers, lacking dense support, are assigned to the noise label. Pre-splitting by orientation is essential: it prevents a tread and the riser directly beneath it—which are spatially adjacent—from merging into a single density-connected cluster. Each resulting cluster is then confirmed by its aggregate slope, i.e. the verticality $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}|$ of its fitted plane, and labelled tread, riser, or other accordingly. This combination of clustering and region-level geometric confirmation mirrors established point-cloud clustering and region-growing practice [14], [16].

F. Height-Histogram Analysis

The fourth method is the most specialised to the staircase structure and ignores normals entirely. Because every tread of an ideal staircase lies at a discrete height ih , the marginal distribution of the z coordinate concentrates into sharp peaks at those levels, separated by low-density bands corresponding to the riser sweeps. We therefore build a one-dimensional histogram of z over the whole cloud and detect its dominant peaks; the points falling within a window of half-width 0.05 m around each peak are labelled tread, while points in the intervening height bands are labelled riser, and the remainder become other. On the benchmark cloud this recovers four tread levels at $z \approx \{0.00, 0.18, 0.36, 0.54\}$ m, matching the generative parameters. The method is extremely fast and requires no neighbourhood computation, but it presumes a regular, axis-aligned staircase and, as the evaluation shows, its height bands tend to over-include riser points near each tread level.

G. Implementation and Parameters

All methods were implemented in Python with NumPy and scikit-learn; RANSAC and the normal-based gates are custom, while DBSCAN and the nearest-neighbour search reuse standard library routines. Table I consolidates every parameter for reproducibility. The four methods share an asymptotic cost dominated by the k -nearest-neighbour normal estimation ($O(M \log M)$ with a k -d tree for M points); RANSAC adds a term linear in the iteration budget, DBSCAN a near-linear neighbourhood query, and the height-histogram

TABLE I
HYPER-PARAMETERS HELD FIXED ACROSS ALL EXPERIMENTS.

Symbol	Value	Description
w, d, h	2.5, 0.30, 0.18 m	width, tread depth, step height
N	4	number of steps
σ_z, σ_y	0.02, 0.02 m	tread / riser Gaussian noise
jitter	0.005 m	isotropic LiDAR-like jitter
outliers	4%	uniform outlier fraction
τ	0.035 m	RANSAC inlier threshold
min. inliers	2,000 / 300	RANSAC tread / riser plane
k	50	PCA normal neighbours
gate (RANSAC)	0.93 / 0.25	horiz. / vert. $ \hat{\mathbf{n}}^\top \hat{\mathbf{z}} $
gate (PCA)	0.80 / 0.50	tread / riser $ \hat{\mathbf{n}}^\top \hat{\mathbf{z}} $
ε	0.08 m	DBSCAN radius
window	0.05 m	histogram peak half-width

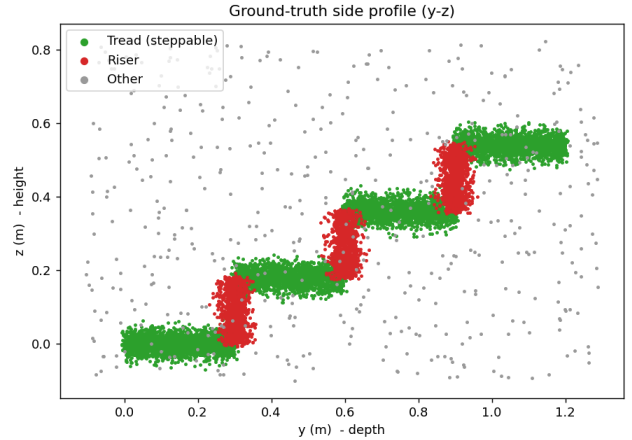


Fig. 1. Side profile (y - z) of the ground-truth cloud: green = tread, red = riser, grey = other.

is $O(M)$, making it by far the cheapest. All values were held fixed across the experiments reported in Section IV.

IV. EVALUATION

This section reports the empirical comparison of the four classical segmentation methods on the parametric staircase point cloud. We first describe the experimental setup and the metrics used (A), then present and interpret the quantitative results (B), study robustness under increasing height noise (C), and finally discuss the safety implications for stair-following robots (D).

A. Experimental Setup and Metrics

All experiments were performed on the synthetic staircase introduced in Section III, with $N = 4$ steps, width $w = 2.5$ m, tread depth $d = 0.30$ m, and step height $h = 0.18$ m. The generated cloud contains 18,333 points, partitioned into 12,800 tread points (label 0), 4,800 riser points (label 1), and 733 “other” points (label 2) arising from edges, LiDAR jitter, and the injected outliers. Figure 1 shows the y - z side profile of the ground-truth cloud, in which the alternating horizontal treads and vertical risers of the staircase are clearly visible together with the sparse off-surface clutter.

TABLE II

TREAD-CLASS SEGMENTATION METRICS (ONE-VS-REST). ACC IS ONE-VS-REST TREAD ACCURACY, DISTINCT FROM THE THREE-CLASS ACCURACY QUOTED IN THE TEXT. BEST PER COLUMN IN BOLD.

Method	Acc	Prec	Rec	F1	IoU
RANSAC	0.681	0.714	0.904	0.798	0.664
PCA-Normals	0.901	0.954	0.901	0.927	0.864
DBSCAN-Slope	0.920	0.936	0.949	0.943	0.892
Height-Histogram	0.827	0.809	0.984	0.888	0.799

Because the steppable surface is the safety-critical class, we evaluate every method on the tread label in a one-vs-rest setting: tread points are treated as positives and all riser and “other” points as negatives. Let TP , FP , FN , and TN denote the per-point true positives, false positives, false negatives, and true negatives. We report five standard quantities, all computed for the tread class in this one-vs-rest setting. The tabulated accuracy $Acc = (TP+TN)/(TP+TN+FP+FN)$ is the fraction of points correctly placed inside or outside the tread class. In the text we additionally quote the overall *three-class* accuracy, defined as the fraction of all points whose full tread/riser/other label is correct; this is a distinct quantity from the one-vs-rest Acc of Table II. Precision $P = TP/(TP+FP)$ is the fraction of predicted tread points that are genuinely treads, and recall $R = TP/(TP+FN)$ is the fraction of true tread points that are recovered. The F1 score is their harmonic mean, $F1 = 2PR/(P+R)$, and the intersection-over-union, or Jaccard index, $IoU = TP/(TP+FP+FN)$, expresses the spatial overlap between the predicted and ground-truth tread regions. For robotic foothold reasoning, precision and IoU are the most informative measures, since they directly quantify how often a non-steppable surface is mistaken for a steppable one.

To guarantee reproducibility, the data generator and every stochastic component of the pipeline (RANSAC sampling, neighbour selection, and the noise draws) were driven by a fixed pseudo-random seed, so the cloud and all reported numbers regenerate deterministically. Method hyper-parameters were held constant across all runs: RANSAC used an inlier threshold $\tau = 0.035$ m, PCA normals were estimated from $k = 50$ neighbours, and DBSCAN used $\varepsilon = 0.08$ m within each orientation group.

B. Results

Table II summarises the tread-class performance of the four methods, and Fig. 2 visualises the corresponding F1 and IoU scores side by side.

The orientation-aware **DBSCAN-Slope** method is the strongest overall, attaining the best F1 (0.943) and IoU (0.892) as well as the highest three-class accuracy (0.909). By pre-splitting the cloud into horizontal and vertical orientation groups and only then clustering spatially, each physical surface is recovered as a single compact cluster while outliers are absorbed into the noise label, yielding a balanced precision (0.936) and recall (0.949). Figure 3 contrasts the DBSCAN-Slope segmentation with the ground truth, showing that the

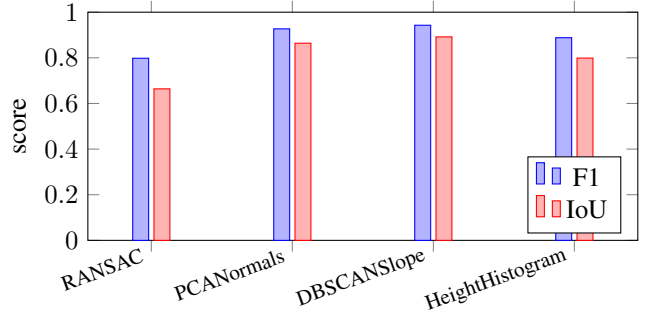


Fig. 2. Per-method tread-class F1 and IoU.

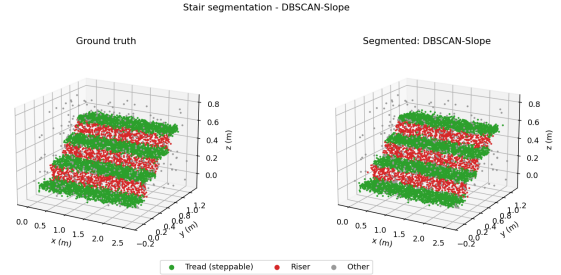


Fig. 3. 3D segmentation by DBSCAN with slope analysis (right) vs. ground truth (left).

individual treads are cleanly separated and that riser bleed-through is minimal.

The **PCA-Normals** method follows closely, with F1 0.927 and IoU 0.864. Because it classifies points directly from the local surface orientation $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}|$, it achieves the highest precision of all methods (0.954): a point is only labelled as tread when its estimated normal is nearly vertical, so very few risers are admitted. Its recall (0.901) is marginally lower than DBSCAN’s because normals estimated near tread–riser edges become unreliable and a fraction of genuine tread points fall outside the horizontal gate. The small gap between PCA-Normals and DBSCAN-Slope confirms that local geometric orientation is an effective and robust cue for separating the two principal surface types.

The **Height-Histogram** method behaves as a high-recall, lower-precision detector. It attains the best recall of all methods (0.984) but a markedly lower precision (0.809), giving F1 0.888 and IoU 0.799. Figure 4 shows the z -histogram, whose four pronounced peaks correspond to the detected tread levels at $z \approx \{0.00, 0.18, 0.36, 0.54\}$ m, in close agreement with the nominal step heights. The method captures almost every tread point because treads concentrate sharply at these levels; however, the horizontal bands it assigns around each peak also enclose riser points whose vertical extent happens to pass through a tread height, which depresses precision.

Plain **RANSAC** plane fitting is the weakest performer, with the lowest precision (0.714), F1 (0.798), IoU (0.664), and three-class accuracy (0.667), despite a high recall (0.904). The dominant failure mode is that the large horizontal planes

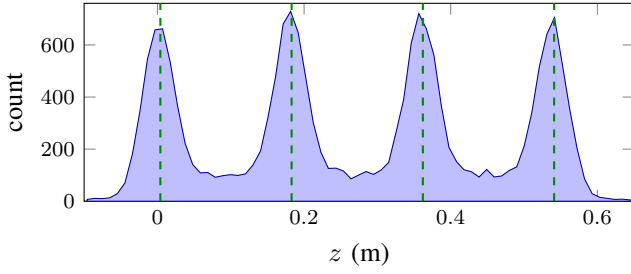


Fig. 4. Height histogram of z ; dashed lines mark the four detected tread levels.

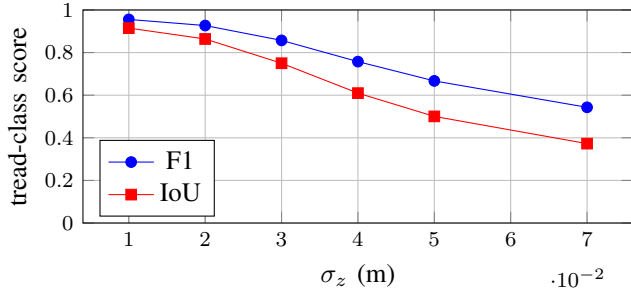


Fig. 5. Tread-class F1 and IoU versus height noise σ_z (PCA-Normals).

RANSAC fits at each tread height also absorb riser points that lie within the inlier band τ at those heights; these mislabelled risers inflate the false-positive count and collapse precision. The contrast with PCA-Normals is instructive: a global planar fit cannot distinguish a riser point that merely happens to sit near a tread plane from a true tread point, whereas a local normal can. This explains why per-point orientation cues, used by both PCA-Normals and DBSCAN-Slope, dominate the purely planar formulation.

C. Noise Sensitivity

To probe robustness, we swept the height-noise standard deviation σ_z from 0.01 m to 0.07 m and re-evaluated the PCA-Normals method, whose accuracy depends most directly on the fidelity of local geometry. We vary σ_z rather than the riser depth jitter σ_y because vertical jitter is the axis that directly erodes the tread/riser orientation contrast, whereas σ_y mainly broadens riser planes without moving tread points across step levels. Figure 5 plots the resulting tread-class F1 and IoU.

Performance degrades monotonically and increasingly steeply as the noise grows. At $\sigma_z = 0.01$ m the method is near-ideal (F1 0.956, IoU 0.915), and it remains strong through $\sigma_z = 0.02$ – 0.03 m. Beyond this point the decline accelerates, falling to F1 0.543 and IoU 0.373 at $\sigma_z = 0.07$ m. The mechanism is geometric: the discriminative cue is the near-vertical tread normal, but as σ_z approaches half the step height ($h/2 = 0.09$ m), vertical jitter on tread points becomes comparable to the vertical separation between adjacent steps. The local neighbourhoods used for normal estimation then mix points from neighbouring treads and the intervening riser, so estimated normals tilt away from vertical, the horizontal

orientation gate misfires, and tread and riser become geometrically indistinguishable. This identifies a practical operating limit: reliable tread segmentation requires the sensor’s per-point height noise to stay well below the step height it must resolve.

D. Limitations and Threats to Validity

Several limitations bound the scope of these conclusions. First, the benchmark is synthetic: a single straight-flight, rectangular staircase with one fixed geometry (w, d, h, N) and an idealised, axis- and gravity-aligned layout. The absolute scores should therefore be read as *relative* method rankings rather than field-deployable accuracies. Second, the sensor model is a nominal Gaussian jitter plus uniform outliers; it omits occlusion, missing returns, incidence-angle effects, multi-path, and the structured scan patterns of a real LiDAR. Third, the height-histogram method explicitly assumes a regular, gravity-aligned staircase and would not transfer to spiral, open-riser, or unaligned scans without modification. Fourth, every hyper-parameter (τ, k, ε , the orientation gates, and the histogram window) was tuned once on this cloud, and the sensitivity study varies only σ_z for a single method; we therefore make no claim of cross-configuration parameter robustness. These constraints motivate the real-sensor validation identified in Section V.

E. Safety Discussion for Stair-Following Robots

For a stair-climbing or stair-following robot, the value of these results lies less in the average score than in the failure modes they expose. The best method reaches IoU ≈ 0.89 , which is sufficient to support *coarse* foothold planning: the extracted tread regions overlap the true steppable surfaces strongly enough for an elevation-map-based planner to place candidate footholds on the correct steps, consistent with practice in legged and wheeled terrain perception [7], [8]. However, no method reaches unit precision, and the dominant errors are precisely the dangerous ones: riser points are occasionally labelled as steppable. In a closed control loop such false positives can lead a controller to commit a foothold to a vertical face, the failure mode that motivates dedicated stair perception in the first place [5].

We therefore recommend that a deployed system treat the raw per-point labels as a hypothesis rather than a command. Three mitigations follow directly from our findings. First, since the methods fail in complementary ways—RANSAC and the histogram over-include risers while PCA-Normals is conservative—*ensemble voting* across methods can suppress method-specific false positives and combine the high recall of the histogram with the high precision of the normal-based gate. Second, because mislabelling concentrates at tread–riser boundaries, a *tread-edge erosion margin* that shrinks each accepted tread region inward before foothold selection trades a small amount of usable surface for a sharp reduction in the chance of stepping onto an edge or riser. Third, given the strong noise sensitivity established above, *temporal verification*—accumulating evidence across consecutive scans

before committing a foothold—guards against transient mislabels caused by momentary high-noise frames. Together these measures convert a segmentation that is accurate on average into one whose residual errors are biased toward the safe side, which is the appropriate design objective for robots that physically traverse stairs.

V. CONCLUSION

We presented a parametric, fully labeled staircase point-cloud benchmark and an apples-to-apples comparison of four classical geometric methods—RANSAC plane fitting, PCA normal analysis, DBSCAN with slope confirmation, and height-histogram banding—under a unified three-class formulation of treads, risers, and other points. On the steppable tread class, orientation-aware DBSCAN with per-cluster slope confirmation was the strongest method, reaching an F1 of 0.943 and an IoU of 0.892, with PCA-normals a close second. We emphasise that these findings are established on a single synthetic staircase under a controlled noise model and should be read as relative method rankings pending real-sensor validation. RANSAC was the weakest, primarily because horizontal planes fitted at tread heights absorb riser points and depress precision to 0.714, whereas the height-histogram attained the highest recall but conflated riser bands with treads. A noise-sensitivity sweep on the PCA-Normals method showed its performance is robust at realistic jitter but degrades sharply as the height noise approaches half the step height, with tread F1 falling from 0.956 at $\sigma_z = 0.01$ m to 0.543 at $\sigma_z = 0.07$ m. From a safety standpoint, an IoU near 0.89 is adequate for coarse foothold planning, but because no method reached unit precision, residual mislabeling of risers as steppable remains a hazard for stair-following robots and assistive devices, motivating conservative tread-edge margins and downstream verification. Future work will pursue ensemble fusion of the complementary geometric cues to recover precision, validation on real LiDAR staircases beyond the synthetic generator, and a controlled comparison against learned point-cloud segmenters [6], [17] under low-label regimes, where the reproducibility and interpretability of classical methods are most valuable.

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